

HARSHIT GOYAL

VLSI & Semiconductor Engineer — B.Tech Electrical Engineering, IIT Delhi

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EDUCATION

Indian Institute of Technology Delhi

2022 – 2026

B.Tech, Electrical Engineering; CGPA: 7.513

EXPERIENCE

Software Development Engineer Intern

Jul 2026 – Present

Samsung R&D Institute

Delhi

- Training and fine-tuning Vision-Language Models (VLMs) to build a custom multimodal solution addressing an unmet product need.

ASIC Synthesis Intern

May 2025 – Jul 2025

Cadence Design Systems

Noida

- Drove Verilog RTL-to-Pre-CTS synthesis flows for multiple design blocks in Cadence Genus and Innovus, taking designs from elaboration and logic synthesis through floorplanning and placement.
- Analyzed setup/hold timing slack, area, power, and standard-cell (library-cell) utilization at each stage, comparing QoR across runs to isolate bottlenecks and drive constraint and optimization tuning.
- Built a Python netlist-parsing tool extracting per-instance cell-utilization and QoR metrics from mapped netlists, and migrated legacy TCL synthesis automation to Python — cutting execution time and replacing manual report inspection.
- Iterated on SDC constraints and synthesis strategies across successive runs, correlating post-synthesis estimates against post-placement results to close timing and area gaps.

Technology Consulting Intern

Jun 2024 – Jul 2024

PwC India

Gurugram

- Digitized financial-service workflows for a PACS cooperative network serving 130M+ farmers and automated the underlying data processing in Python, cutting end-to-end processing time by 15%.

Research Intern

Dec 2023 – Jan 2024

Cape Peninsula University of Technology

Cape Town

- Researched AI-enabled smart-microgrid architectures and embedded/edge deployment constraints, building a structured evaluation framework and authoring a 32-source technical report to guide adoption.

PROJECTS

Codebase Intelligence Platform – Graph-Based Static Analysis

github.com/goyal-harshit/codebase-intelligence-platform

tree-sitter · ArcadeDB · ChromaDB · FastAPI

- Ingests any Git repository, parsing source with tree-sitter into a language-agnostic dependency graph (ArcadeDB) plus a semantic vector index (ChromaDB) for structural and natural-language queries.
- Detects god objects, dead code, and circular dependencies via call/import graph traversal; parsed a 24-file Flask codebase in 0.28s, extracting 388 functions and 53 classes.
- Answers plain-English code questions through hybrid retrieval, compiling natural language to Cypher with semantic vector fallback and re-parsing only changed files on re-index.

QuantAI – Quantitative Investment Research Platform

github.com/goyal-harshit/quant-analyzer

FastAPI · Next.js · PostgreSQL/TimescaleDB · Docker

- Built a full-stack quant research platform for NSE/BSE equities with a multi-factor screener and an event-driven backtester over TimescaleDB time-series storage.
- Added a portfolio tracker and a RAG-grounded AI assistant with a swappable LLM backend; containerized with Docker and shipped GitHub Actions CI/CD.
- Put every external data source behind a circuit breaker and rate limiter with fallback chains, exposing Prometheus metrics and per-source health for observability.

AI/ML Fantasy Cricket Predictor – 9th Nationally, FIFS Gameathon 2025

Google Cloud & Dream11

- Built an automated CricSheet pipeline with match-wise feature engineering, then trained a Random Forest / Gradient Boosting / PyTorch player-embedding ensemble — placing 9th nationally.
- Modeled player form, matchup, and venue effects, then selected final teams under budget and role constraints via constrained optimization over predicted scores.

TECHNICAL SKILLS & ACHIEVEMENTS

Digital & VLSI: Verilog, RTL Design, Logic Synthesis, Static Timing Analysis (STA), CMOS, CPLD/FPGA

EDA & Tools: Cadence Genus, Cadence Innovus, LTSpice, GVim, Git/GitHub Actions, Docker

Programming: Python, TCL, C++, MATLAB, Verilog

ML & Software: PyTorch, TensorFlow, FastAPI, scikit-learn

Achievements: JEE Advanced 2022 – AIR 530 (GE) of 250,000+; JEE Main 2022 – 99.74 percentile of 1,000,000+